

# Origin-Destination Matrix Estimation from Traffic Measurements

Experiments report — Tehran, District 2 (1,897 road links) · SUMO + Graph Neural Networks

**Abstract.** We study the inverse traffic problem: recovering an origin-destination (OD) trip matrix from road-level traffic measurements. Across four experiments we establish one central result — **the OD matrix is only as recoverable as the measurement is informative**, and the bottleneck is the *data*, not the model. Aggregate link counts leave the OD under-determined (cell-correlation 0.14 for an optimal linear inverse); **junction turn counts** make the zone marginals recoverable (correlation  $\approx 0.90$ ) and, combined with a doubly-constrained gravity reconstruction, yield **OD cell-correlation 0.78** at 36 zones. Real-world **travel-time** data (Neshan) is the weakest signal of all (ceiling 0.06): it can be used to *calibrate* a SUMO scenario that reproduces observed conditions, but it cannot identify the true OD. We report results at 36, 9 and 13 zones, a real-data calibration pipeline, and the engineering defects found along the way.

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## 1. Objective and the inverse problem

The forward pipeline (`od_generator/`, generalised in `src/odpipe/`) samples an OD matrix, routes it through SUMO, and records per-edge traffic. We build the **inverse**:

per-edge SUMO measurements  $\rightarrow$  **GNN**  $\rightarrow$  OD matrix (origin  $\times$  destination over TAZ zones).

This is fundamentally **ill-posed**: many different OD matrices produce nearly the same link measurements. Every result below follows from how much of the OD a given measurement actually constrains.

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## 2. Method

The recipe that works, used identically in every experiment:

1. **Road line-graph** — nodes = the 1,897 road edges (features: length, lanes, speed limit, plus measured flow / speed / density / travel time and turn aggregates); edges = downstream connectivity; **edge weights = junction turn counts**.
2. **Predict the recoverable quantity** — a GNN (`ODMarginalGNN`, `GraphConv`) pools the graph into per-zone vectors and predicts **production/attraction marginals**, trained with a per-zone *standardised MSE* (a  $\log_{1p}$ +Huber loss under-disperses the marginals and collapses the reconstruction).
3. **Reconstruct the matrix** — doubly-constrained gravity / **Furness (IPF)** from the predicted marginals with a distance-decay deterrence.

Splits are stratified by demand tier so every split covers all demand regimes. Metrics: marginal correlation, OD cell-correlation (pooled, off-diagonal), total-flow error, and **GEH** on re-simulated link flows.

### 3. Experiment 1 — What measurement identifies the OD? (the central result)

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We fixed the model class (ridge) and varied only the *input signal*, on synthetic data where the truth is known, under a realistic 77 % edge coverage.

| input signal                                | zone-marginal corr | reconstructed OD cell-corr |
|---------------------------------------------|--------------------|----------------------------|
| static only (no traffic)                    | 0.00               | 0.02                       |
| <b>travel time</b> (what Neshan provides)   | 0.25               | <b>0.06</b>                |
| speed                                       | 0.24               | 0.05                       |
| link flow (counts)                          | 0.36               | 0.14                       |
| junction turn counts (node-aggregated)      | 0.46               | 0.19                       |
| all dynamic features                        | 0.50               | 0.20                       |
| <i>ceiling: gravity from TRUE marginals</i> | 1.00               | <b>0.87</b>                |

Two conclusions drive the whole project. **(a)** No model — GNN, VAE or ridge — can recover the OD from travel time alone; the information is not there. **(b)** The gravity reconstruction from *true* marginals reaches 0.87, so **predicting marginals + gravity is near-optimal**; the remaining task is to predict marginals well, which turn counts enable.

### 4. Experiment 2 — 36 zones (turn counts + marginal GNN)

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Re-simulating each scenario to capture junction turn counts, and feeding them as GNN edge weights, lifts marginal correlation from  $\approx 0.5$  to  $\approx 0.90$ .

| approach (36 zones, test n=750)            | OD cell-corr | total-flow err | RMSE        |
|--------------------------------------------|--------------|----------------|-------------|
| direct OD-cell regression (baseline)       | 0.06         | 0.44           | 6.78        |
| marginal GNN + gravity, <b>link counts</b> | 0.435        | 0.49           | 6.78        |
| marginal GNN + gravity, <b>turn counts</b> | <b>0.784</b> | <b>0.132</b>   | <b>4.08</b> |

Marginals: production  $r = \mathbf{0.904}$ , attraction  $r = \mathbf{0.916}$ . The direct-regression baseline confirms the under-determination; the turn-count model is a 13× improvement in cell-correlation over it.

### 5. Experiment 3 — 9 zones, and 36 vs 9

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The same recipe re-run at a coarser resolution (TAZ grid width 3000 m  $\rightarrow$  9 zones, 2,000 samples).

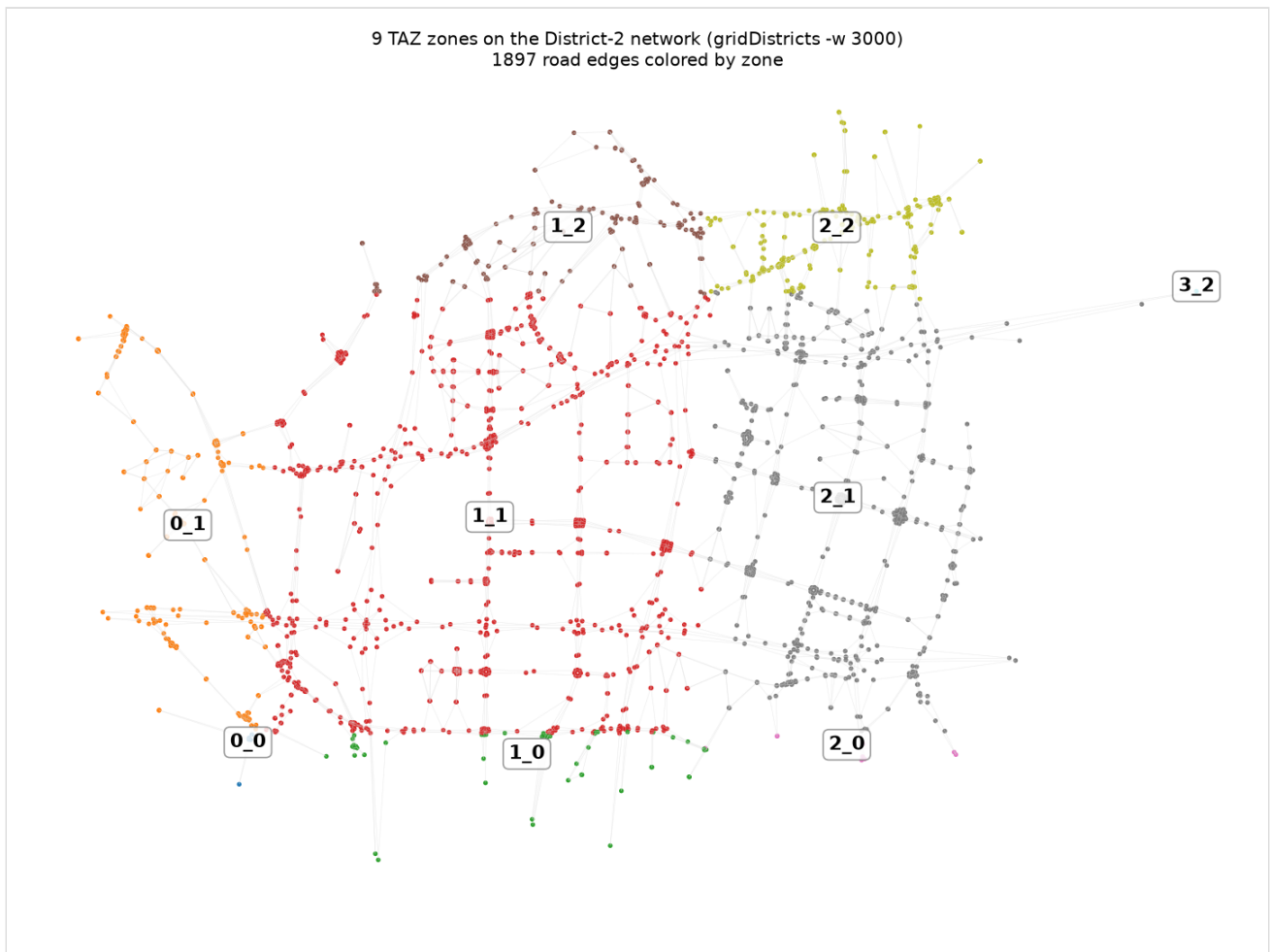


Figure 1 — The 9 TAZ zones on the District-2 network. Note the imbalance: the central zones dominate while edge cells are tiny — a recurring source of noisy marginals.

| metric                         | 36 zones             | 9 zones       |
|--------------------------------|----------------------|---------------|
| OD cells (off-diagonal)        | 1,260                | 72            |
| training samples               | 3,500                | 2,000         |
| production / attraction corr   | <b>0.904 / 0.916</b> | 0.816 / 0.800 |
| OD cell-corr                   | <b>0.784</b>         | 0.660         |
| total-flow error               | <b>0.132</b>         | 0.186         |
| gravity ceiling                | ~0.86                | 0.884         |
| <b>GEH &lt; 5 (link flows)</b> | not scored           | <b>94.4 %</b> |

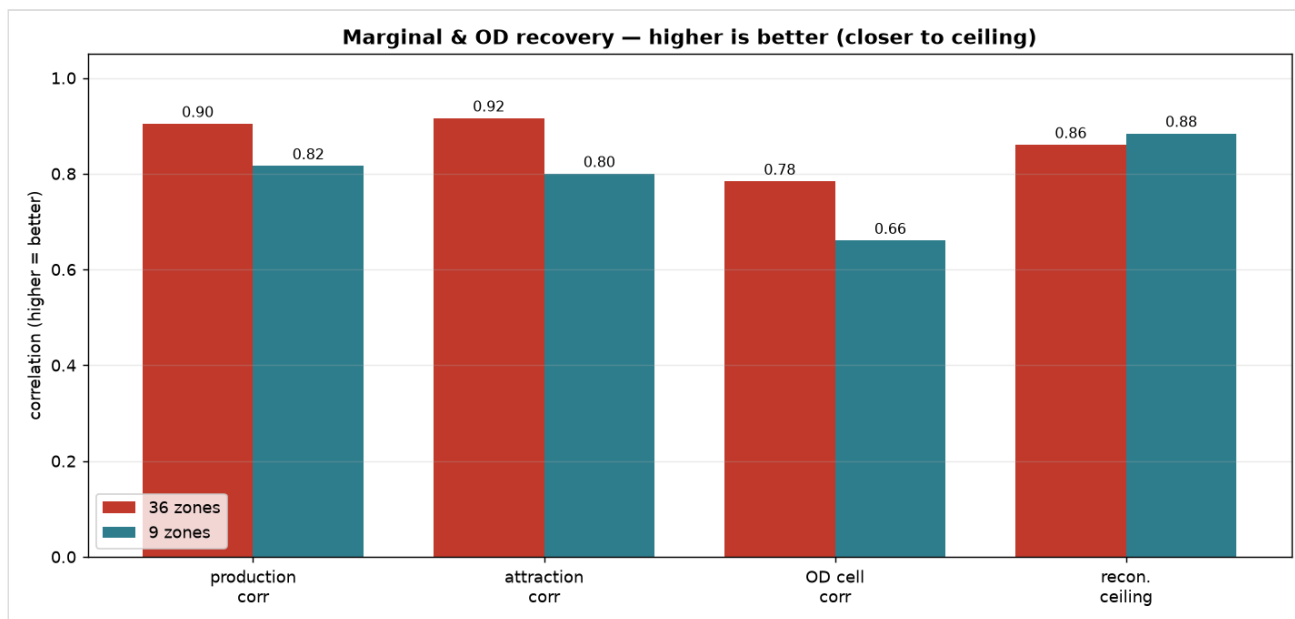


Figure 2 — Marginal and OD recovery at both resolutions, against the gravity ceiling. Both models sit close to their ceiling: the limit is information, not capacity.

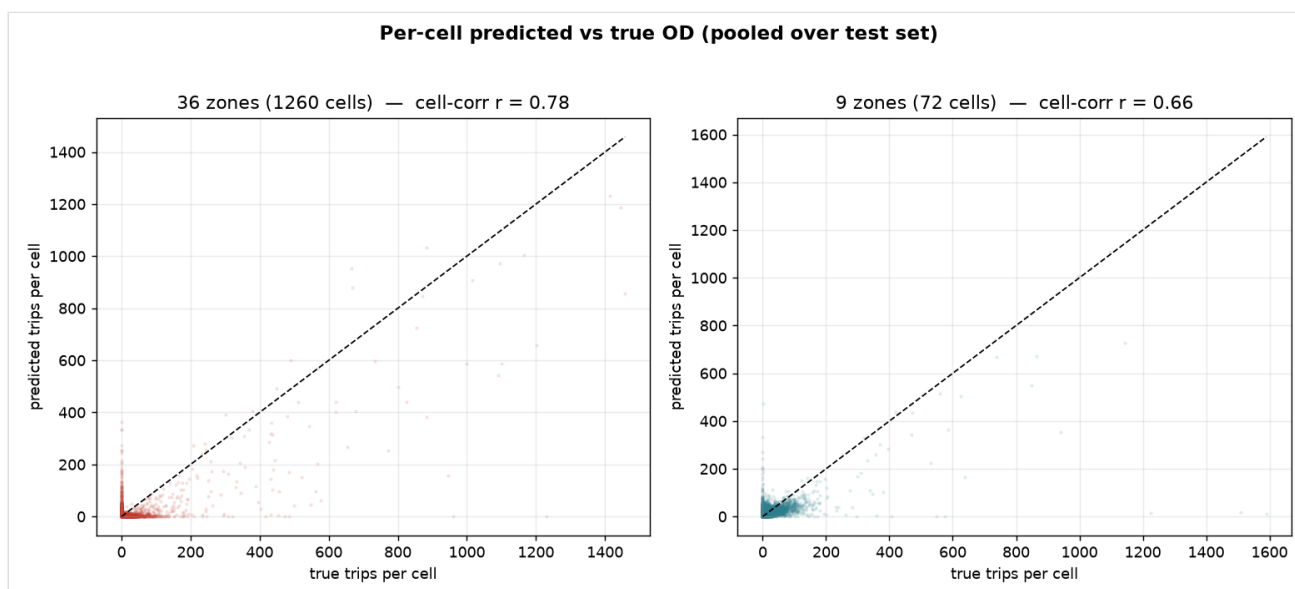


Figure 3 — Per-cell predicted vs true OD, pooled over the test set.

**Finer is not harder here.** The 36-zone model scores *higher* on every correlation metric despite having 17× more cells — largely because it was trained on 1.75× more data, and because coarse grids still contain tiny, near-empty zones that add noise. Conversely the 9-zone OD is the most **simulation-faithful**: re-simulating it reproduces observed link flows with GEH < 5 on 94 % of links. Raw RMSE is *not* comparable across resolutions (the same trips packed into 72 vs 1,260 cells make 9-zone per-cell values ~6× larger).

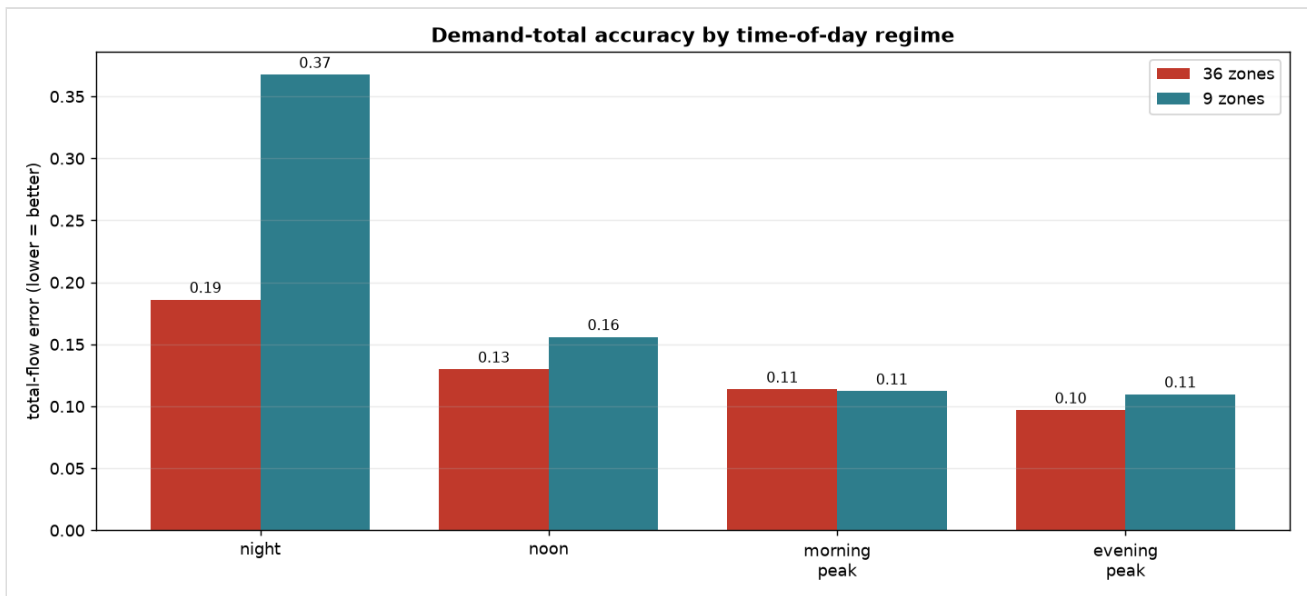


Figure 4 — Accuracy by demand regime. Both improve with demand; low-demand (night) is the hardest, where few trips give little signal.

## 6. Experiment 4 — General pipeline: 13 zones at high demand (mesoscopic)

To test generality we rebuilt the pipeline config-driven (`src/odpipe/`, nothing hard-coded to a zone count) and ran it at **TAZ width 2,000 m → 13 zones** with a **high demand range of 2,000-30,000 trips/hour**, simulated with SUMO's **mesoscopic** model (~2 s/sample — microsimulation is impractical at this demand).

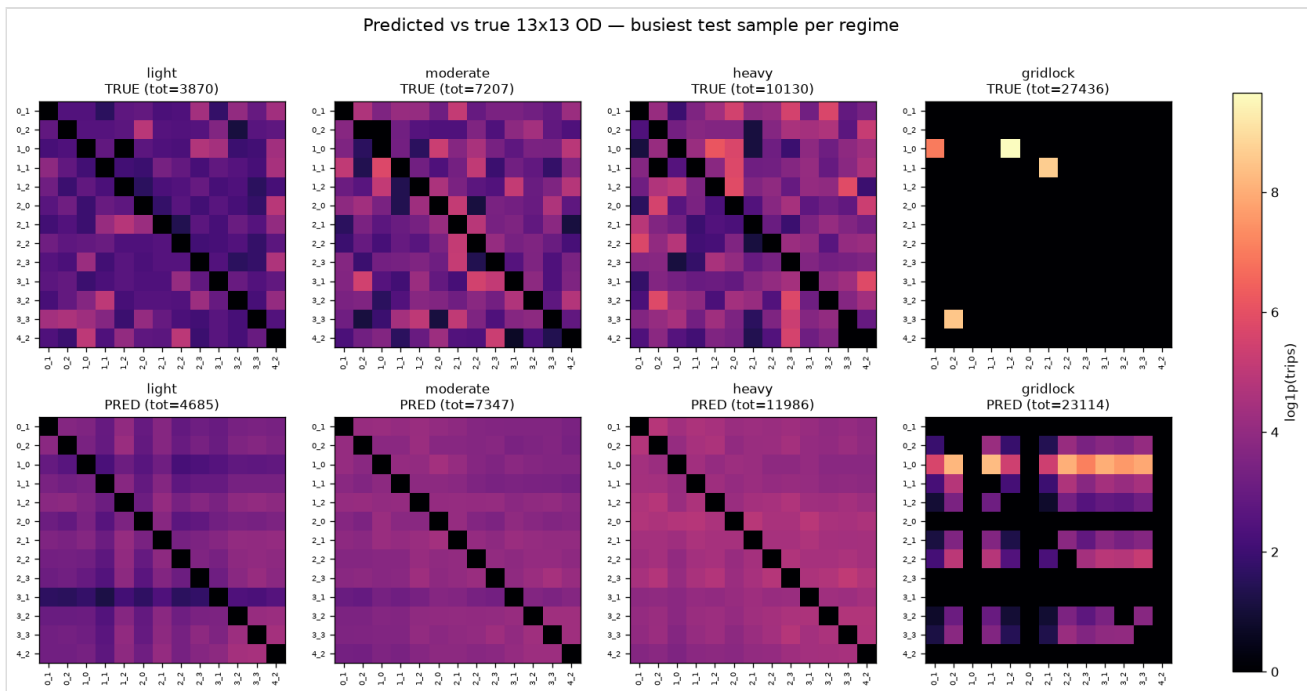


Figure 5 — Predicted vs true 13x13 OD, busiest test sample per demand tier. The blank rows/columns in the predictions are the three tiny zones (2-3 edges each) produced by the 2,000 m grid: they carry almost no traffic, so their marginals are unlearnable.

| 13 zones, high demand (smoke run: 160 samples, test n=24) | value         |
|-----------------------------------------------------------|---------------|
| production / attraction corr                              | 0.749 / 0.390 |
| OD cell-corr                                              | 0.164         |
| total-flow error                                          | 0.270         |
| gravity ceiling                                           | 0.825         |
| GEH < 5                                                   | 48.3 %        |

These are **shakedown numbers from a deliberately tiny dataset** (160 samples, to validate the plumbing) and should not be compared with the fully-trained experiments above. Two real lessons do carry: the **grid width matters more than the zone count** — a width that produces near-empty zones (attraction  $r = 0.39$  here) caps achievable accuracy; and **mesoscopic simulation is what makes high-demand study feasible** at all.

## 7. Experiment 5 — Real data: OD calibration from Neshan travel times

The operational goal is a SUMO scenario that reproduces *observed* traffic so that road changes can be simulated. The only real observable is **per-segment travel time** — by Experiment 1, the weakest signal (ceiling 0.06). We therefore do **not** learn the OD; we **calibrate** it with SUMO in the loop.

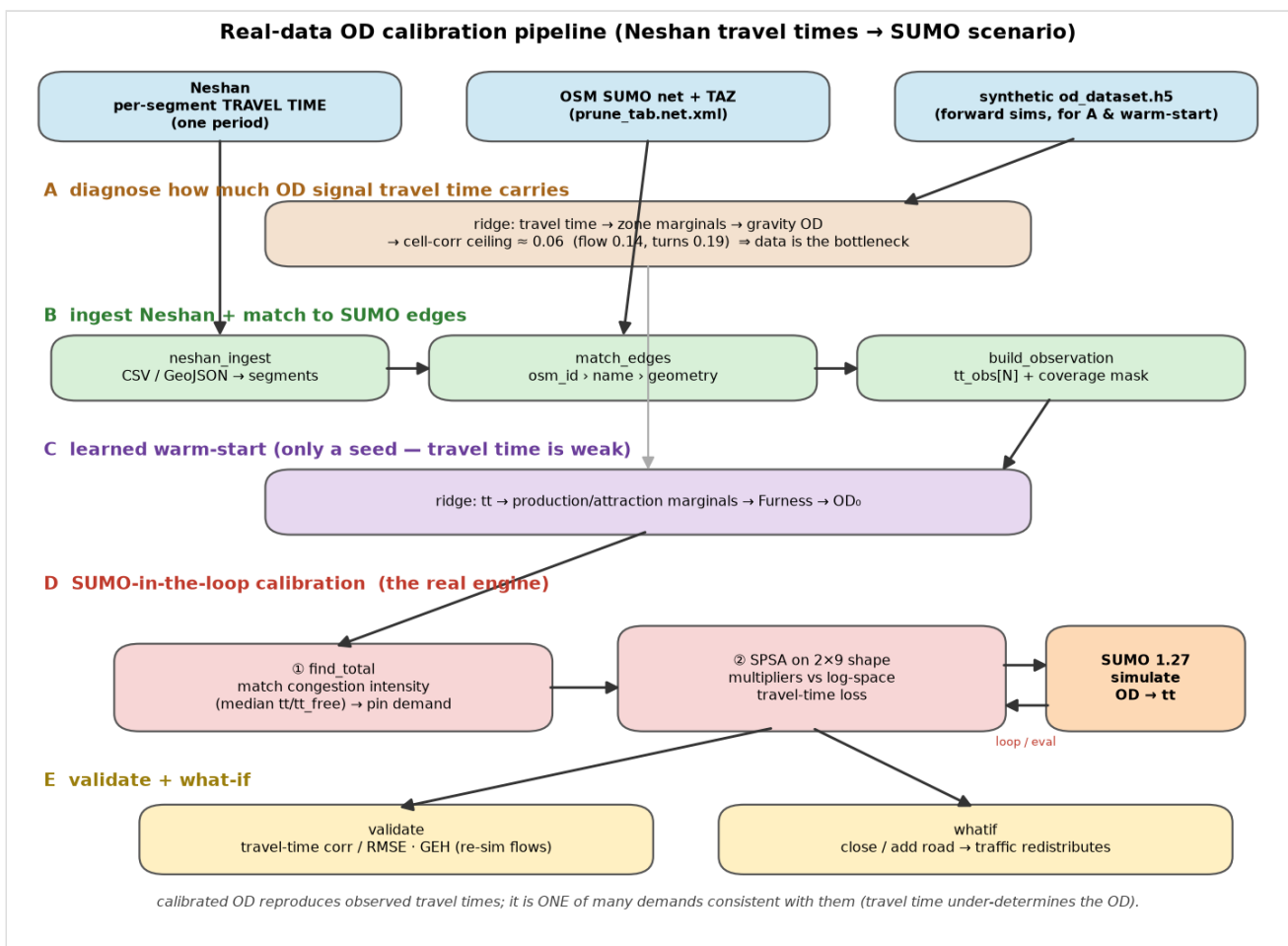


Figure 6 — The real-data calibration pipeline. A ridge model gives only a warm start; the engine is a SUMO-in-the-loop search that matches observed travel times.

The demand is split into the parts travel time can inform: **total demand** is pinned by matching the overall congestion intensity, then **SPSA** (gradient-free) optimises the spatial shape against a log-space travel-time loss on the observed edges.

On a self-consistent benchmark (a known 1,906-trip evening-peak OD, 398 observed edges):

|                                    | warm start | calibrated                  |
|------------------------------------|------------|-----------------------------|
| travel-time correlation            | 0.677      | <b>0.722</b>                |
| travel-time log-RMSE               | 1.452      | <b>1.185</b>                |
| observed edges actually reproduced | 58 %       | <b>77 %</b>                 |
| SPSA loss (30 iterations)          | 2.110      | <b>1.412</b>                |
| <b>recovered total demand</b>      | —          | 2,322 vs 1,906 true (+22 %) |
| <b>OD cell-corr vs truth</b>       | —          | <b>0.035</b>                |

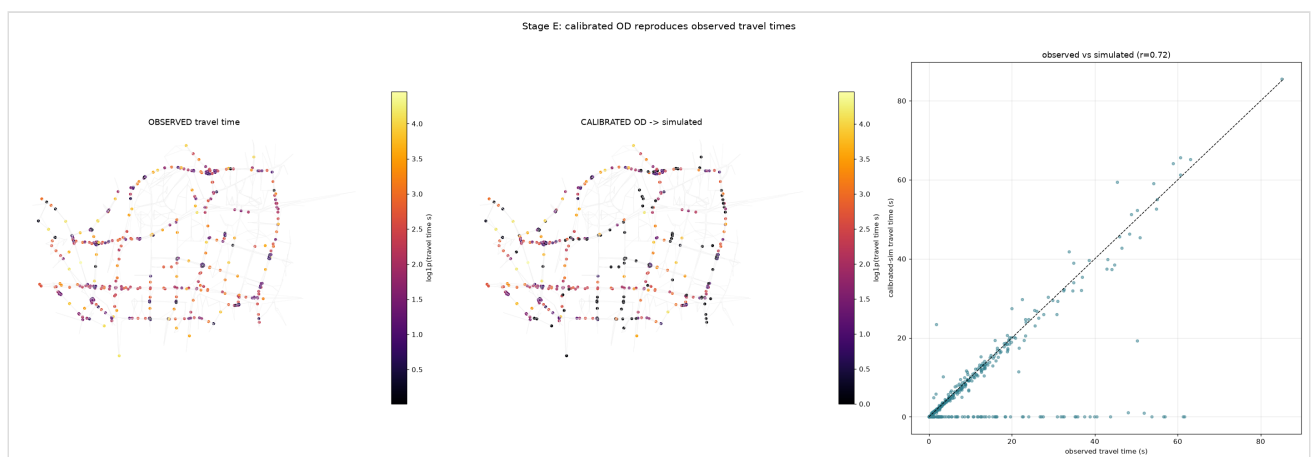


Figure 7 — Observed vs simulated travel times after calibration. Points on the horizontal axis are observed edges the calibrated demand leaves empty — a direct symptom of the non-uniqueness.

**The honest result.** We can recover the *demand magnitude* to ~20 % and reproduce observed travel times, **but not the OD itself** (cell-corr 0.035) — exactly as Experiment 1 predicts. Travel time lets you fit *either* the demand level *or* the per-edge pattern; forcing a tighter per-edge match (travel-time corr 0.90) requires inflating demand 3x. The calibrated matrix is therefore **one of many** demands consistent with today's data.

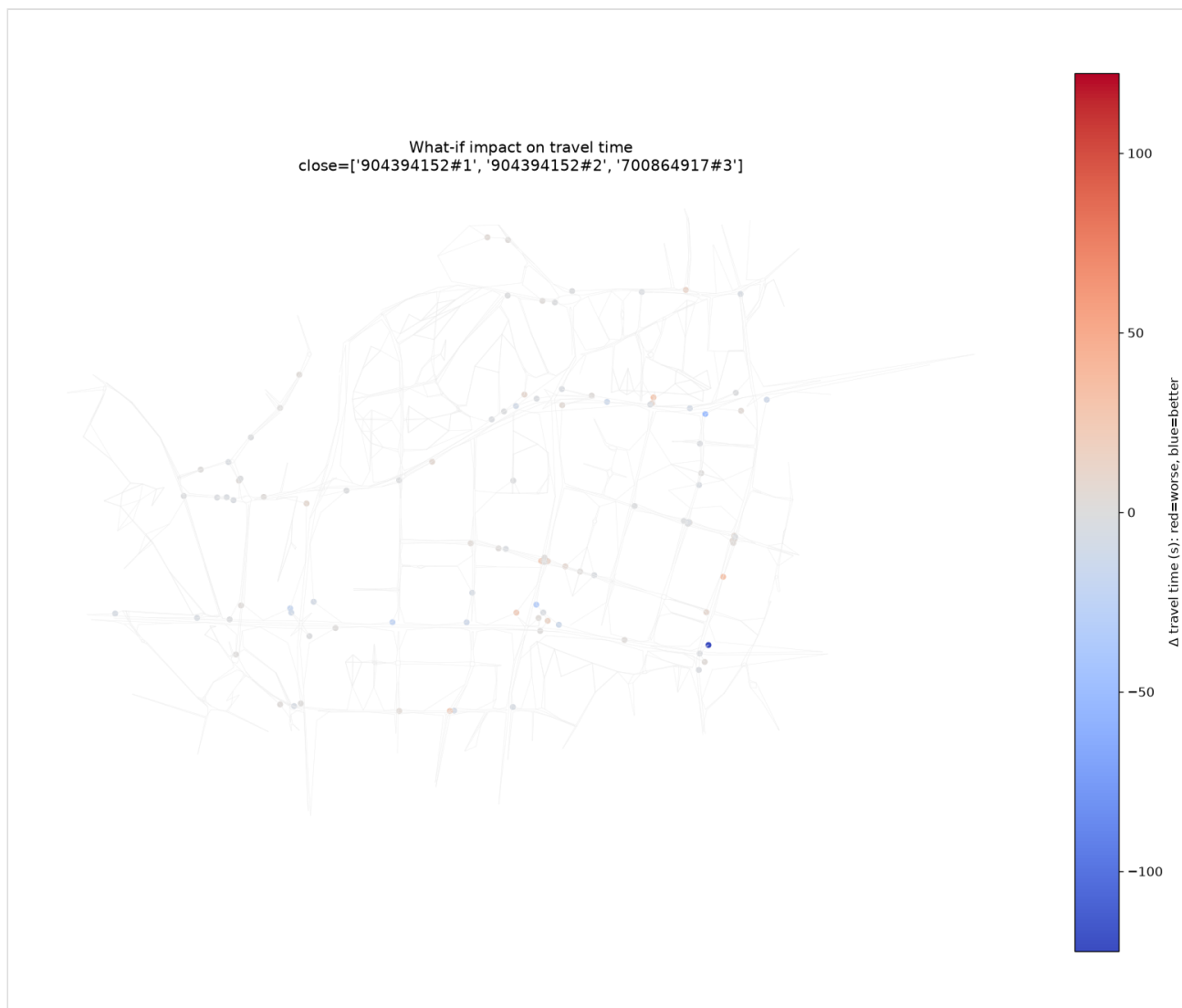


Figure 8 — What-if: closing a three-edge corridor with the calibrated demand held fixed. Travel time changes on 104 edges — relieved links in blue, detour links in red. This is the end goal working; its reliability is highest for moderate, local changes.

## 8. Engineering findings

Defects found and fixed — each one materially affected results:

| finding                                                                                                                                                                                                            | impact                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Route files must be globally sorted by departure.</b> SUMO streams route files and <i>silently drops</i> any vehicle departing before the previous one. Trips were written per OD pair, so most were discarded. | Simulations ran on a fraction of the intended demand (sparse traffic, under-counted congestion, biased calibration). Fixed by sorting all trips by depart time. |
| <b>Race in parallel generation.</b> Workers shared a temp directory keyed by <code>idx % n_workers</code> , so a freed worker could overwrite a running one's trip files.                                          | ~40 % of samples failed. Fixed with a unique directory per sample → 0 failures.                                                                                 |
| <b>SUMO version mismatch.</b> System SUMO 1.18 cannot parse this network (built with 1.26).                                                                                                                        | Pinned to the <code>eclipse-sumo 1.27</code> wheel and resolve binaries from it.                                                                                |



| finding                                                                                                                           | impact                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Marginal loss choice.</b> log1p + Huber under-disperses the marginals (std-ratio 0.35), collapsing the Furness reconstruction. | Per-zone standardised MSE restores dispersion; cell-corr 0.44 → 0.78.         |
| <b>Long processes are killed (~45 s).</b>                                                                                         | All training and calibration are checkpointed and resumable (- -max-seconds). |

## 9. Conclusions

1. **The measurement, not the model, is the bottleneck.** Ranked by OD identifiability: turn counts > link counts > speed  $\approx$  travel time. Adding model capacity or synthetic data cannot cross this ceiling.
2. **Predict marginals, reconstruct with gravity.** The full OD is under-determined; the marginals are not. This reframing plus turn counts takes cell-correlation from 0.06 to **0.78**.
3. **Choose resolution by purpose.** 36 zones for spatial detail and the best correlations; 9 zones for a compact, simulation-faithful OD (GEH < 5 on 94 % of links). Avoid grid widths that create near-empty zones.
4. **For the real network, calibrate — don't learn.** Travel time cannot identify the OD, but SUMO-in-the-loop calibration produces a scenario that reproduces observed conditions and supports what-if road changes, with quantified uncertainty.

**Recommended next step:** obtain any *count-like* measurement (junction turn counts or link volumes) for the real network. That single change moves the real-data problem from the 0.06 regime to the 0.78 regime, and makes SUMO's own `routeSampler/cadyts` calibration directly applicable.

## 10. Reproducibility

All hyperparameters and paths live in `configs/*.yaml`; seeds are fixed; CUDA is auto-selected with CPU fallback.

```
conda activate traffic

# general pipeline (any zone count / demand range) – see README_odpipe.md
python -m src.odpipe.taz                # TAZ at configs/odpipe.yaml -> taz.width
python -m src.odpipe.graph
python -m src.odpipe.generate --workers 8    # mesoscopic, resumable
python -m src.odpipe.build_packs
python -m src.odpipe.train_eval --mode train --device cuda --max-seconds 25
python -m src.odpipe.train_eval --mode eval  --device cpu
python -m src.odpipe.infer --max-trips --sumo-gui --which both --meso
python -m src.odpipe.visualize

# real-data calibration from Neshan travel times – see README_realworld.md
python -m src.realworld.diagnose_traveltime # the identifiability table (Experiment 1)
python -m src.realworld.calibrate --max-seconds 300
```

```
python -m src.realworld.validate  
python -m src.realworld.whatif --close-edges <e1>,<e2>
```